

Amar Telidji University Laghouat

Department of Computer Science

Level: 4th Year Engineering

Course: Advanced Network

Lab 03: Wireshark Lab (HTTP)

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1. The Basic HTTP GET/response interaction:

No.	Time	Source	Destination	Protoc	Leng	Info
21	2.198144	10.131.30.61	128.119.245.12	HTTP	441	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
27	2.399641	128.119.245.12	10.131.30.61	HTTP	547	HTTP/1.1 200 OK (text/html)

Q1. Http version:

```
Hypertext Transfer Protocol
  ▾ GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /wireshark-labs/HTTP-wireshark-file1.html
    Request Version: HTTP/1.1
```

From the Request Line of the first HTTP GET message, the browser is using **HTTP/1.1**

```
Hypertext Transfer Protocol
  ▾ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
```

Similarly, the Status Line of the server's response shows that the web server is also operating over **HTTP/1.1**

Q2. Accepted Languages:

```
Accept: text/html,application/xhtml+xml
Accept-Language: en-US,en;q=0.5\r\n
Accept-Encoding: gzip, deflate\r\n
```

Within the Request Headers of the HTTP GET message, the field indicates that the browser can accept content in **U.S. English (en-US)** as its preferred language, followed by **general English (en)** with a slightly lower preference (**q=0.5**).

Q3. IP addresses:

```
Source Address: 10.131.30.61
Destination Address: 128.119.245.12
```

My IP: **10.131.30.61**

Server IP: **128.119.245.12**

Q4. Status code:

```
  ▾ Hypertext Transfer Protocol
    ▾ HTTP/1.1 200 OK\r\n
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK
```

The status code **200 OK** indicates that the client's request was successfully received, understood, and processed by the server, and that the requested HTML file was returned without errors.

Q5. Last modified date:

```
Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
```

The “Last-Modified” timestamp (**28 Oct 2025**) does not reflect the current date (**12 Nov 2025**) most likely because the capture used a pre-recorded trace or a cached response. The Gaia server normally updates that file every minute, but since the trace or cache preserved an older header, the timestamp appears outdated.

Q6.

```
Content-Length: 128\r\n[Content length: 128]
```

This shows that the server returned **128 bytes** of content in the message body, representing the HTML file's size.

Q7. By inspecting the raw data in the bottom (packet bytes) pane, an additional header can be observed:

```
65 66 6c 61 74 65 0d 0a 43 6f 6e 6e 65 63 74 69 eflate Connection: keep-alive
```

This header is not explicitly displayed in the packet-listing summary but appears in the raw HTTP data. It indicates that the TCP connection should be kept open for possible reuse, reducing the overhead of creating new connections.

2. The HTTP CONDITIONAL GET/response interaction:

No.	Time	Source	Destination	Protocol	Length	Info
14	0.424271	10.131.30.61	128.119.245.12	HTTP	441	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
17	0.617977	128.119.245.12	10.131.30.61	HTTP	791	HTTP/1.1 200 OK (text/html)
26	0.665700	10.131.30.61	128.119.245.12	HTTP	461	GET /favicon.ico HTTP/1.1
29	0.843176	128.119.245.12	10.131.30.61	HTTP	640	HTTP/1.1 301 Moved Permanently (text/html)
51	2.019158	10.131.30.61	128.119.245.12	HTTP	527	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
54	2.202994	128.119.245.12	10.131.30.61	HTTP	368	HTTP/1.1 304 Not Modified

Q8. No, the first GET request does not contain an “If-Modified-Since” header.
It’s a standard GET request since the browser has no cached copy yet.

```
▼ Hypertext Transfer Protocol
  ▶ GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:145.0) Gecko/20100101 Firefox/145.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    Priority: u=0, i\r\n
    \r\n
    [Response in frame: 17]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
```

Q9. Yes, the server explicitly returned the contents of the file.
because the response (Packet 17) shows “HTTP/1.1 200 OK” and includes a Content-Type: text/html and Content-Length header, meaning the file’s content was sent to the browser.

```
▼ Content-Length: 371\r\n
  [Content length: 371]
  Keep-Alive: timeout=5, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=UTF-8\r\n
  \r\n
```

Q10. Yes, in the second GET request (Packet 51), the browser includes the header:

```
Upgrade-Insecure-Requests: 1\r\n
If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT\r\n
If-None-Match: "173-64231b67176b7"\r\n
```

This tells the server the file was last cached at that time, so the browser only wants the file if it has been modified since then.

Q11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET?

The server responded with the status code **304** and phrase **Not Modified** (Packet 54).

```
▼ Hypertext Transfer Protocol
  ▼ HTTP/1.1 304 Not Modified\r\n
    Response Version: HTTP/1.1
    Status Code: 304
    [Status Code Description: Not Modified]
    Response Phrase: Not Modified
    Date: Wed, 12 Nov 2025 20:15:40 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.2.2 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
    Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
    ETag: "173-64231b67176b7"\r\n
    Accept-Ranges: bytes\r\n
    Keep-Alive: timeout=5, max=98\r\n
    Connection: Keep-Alive\r\n
```

Did the server explicitly return the contents of the file?

No, the server did not explicitly return the contents of the file, this code tells the browser its cached copy is still valid, so no file data was retransmitted.

3. Retrieving Long Documents:

Q12. The browser sent **one GET request** for the Bill of Rights, in **packet 9**.

- **GET /favicon.ico (packet 21)**, this one is for the browser icon and should be ignored.

No.	Time	Source	Destination	Protocol	Length	Info
9	0.183585	10.131.30.61	128.119.245.12	HTTP	441	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
16	0.405976	128.119.245.12	10.131.30.61	HTTP	650	HTTP/1.1 200 OK (text/html)
21	0.459252	10.131.30.61	128.119.245.12	HTTP	461	GET /favicon.ico HTTP/1.1
30	0.689021	128.119.245.12	10.131.30.61	HTTP	640	HTTP/1.1 301 Moved Permanently (text/html)

Q13. The response to the GET is in **packet 16**, where we can see:

HTTP/1.1 200 OK (text/html)

16	0.405976	128.119.245.12	10.131.30.61	HTTP	650	HTTP/1.1 200 OK (text/html)
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Q14. Status Code: **200**

Phrase: **OK**

Which means the response sent without errors and the text received fully without any problems.

```
▼ HTTP/1.1 200 OK\r\n
  Response Version: HTTP/1.1
  Status Code: 200
  [Status Code Description: OK]
  Response Phrase: OK
```

Q15. In the TCP section in the Http response (**packet 16**) we can see that it needed **4 packets** to carry the full data:

```
[4 Reassembled TCP Segments (4864 bytes): #13(1424), #14(1424), #15(1424), #16(592)]
[Frame: 13, payload: 0-1423 (1424 bytes)]
[Frame: 14, payload: 1424-2847 (1424 bytes)]
[Frame: 15, payload: 2848-4271 (1424 bytes)]
[Frame: 16, payload: 4272-4863 (592 bytes)]
[Segment count: 4]
[Reassembled TCP length: 4864]
```

4. HTML Documents with Embedded Objects:

Q16. Total GET requests related to the lab is **3 requests**:

P#6: GET /wireshark-labs/HTTP-wireshark-file4.html → **128.119.245.12 (Gaia.cs.umass.edu)**

P#12: GET /pearson.png → **128.119.245.12 (Gaia.cs.umass.edu)**

P#124: GET /8E_cover_small.jpg → **2.56.99.24 (remote image server)**

P#26: GET /favicon.ico → **128.119.245.12 (ignored, automatic browser request)**

No.	Time	Source	Destination	Protocol	Length	Info
6	0.267211	10.131.30.61	128.119.245.12	HTTP	441	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
9	0.559098	128.119.245.12	10.131.30.61	HTTP	1362	HTTP/1.1 200 OK (text/html)
12	0.649774	10.131.30.61	128.119.245.12	HTTP	464	GET /pearson.png HTTP/1.1
15	0.870421	128.119.245.12	10.131.30.61	HTTP	641	HTTP/1.1 301 Moved Permanently (text/html)
26	1.090141	10.131.30.61	128.119.245.12	HTTP	461	GET /favicon.ico HTTP/1.1
33	1.275232	128.119.245.12	10.131.30.61	HTTP	640	HTTP/1.1 301 Moved Permanently (text/html)
124	4.661853	10.131.30.61	2.56.99.24	HTTP	431	GET /8E_cover_small.jpg HTTP/1.1
504	5.555793	2.56.99.24	10.131.30.61	HTTP	300	HTTP/1.1 200 OK (JPEG JFIF image)

Q17. From the timestamps:

- Request for /pearson.png → at **0.649774 s**
- Request for /8E_cover_small.jpg → at **4.661853 s**

The second request occurs about **4 seconds** after the first, This shows that the browser initiated the downloads **serially**.

5 HTTP Authentication:

Q18. The server's first response (**packet 15**) is:

- Status code: **401**
- Response Phrase: **Unauthorized**

It indicates that authentication is required before access is granted to the protected page.

No.	Time	Source	Destination	Protocol	Length	Info
7	0.258363	10.131.30.61	128.119.245.12	HTTP	457	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
15	0.468186	128.119.245.12	10.131.30.61	HTTP	778	HTTP/1.1 401 Unauthorized (text/html)
71	9.627429	10.131.30.61	128.119.245.12	HTTP	516	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
76	9.794769	128.119.245.12	10.131.30.61	HTTP	551	HTTP/1.1 200 OK (text/html)
79	9.852216	10.131.30.61	128.119.245.12	HTTP	477	GET /favicon.ico HTTP/1.1
83	10.030910	128.119.245.12	10.131.30.61	HTTP	640	HTTP/1.1 301 Moved Permanently (text/html)

```
▼ Hypertext Transfer Protocol
  ▼ HTTP/1.1 401 Unauthorized\r\n
    Response Version: HTTP/1.1
    Status Code: 401
    [Status Code Description: Unauthorized]
    Response Phrase: Unauthorized
    Date: Thu, 13 Nov 2025 10:19:09 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.2.2 mod_fcgid/2.3.9
    WWW-Authenticate: Basic realm="wireshark-students only"\r\n
```

Q19. The second GET (**packet 71**) includes the header:

```
► Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcm5=\r\n
```

After receiving the 401 code in the first response in **packet 15**, the browser resends the request with the Authorization header containing the credentials encoded in Base64.

When decoded, the credentials are:

- Username: **wireshark-students**
- Password: **network**

This shows that Basic Authentication is not secure, since anyone capturing the packets with tools like Wireshark can easily decode the credentials.

Conclusion:

In this lab, we used Wireshark to observe the operation of the HTTP protocol and its key features. We analyzed basic GET and response exchanges, conditional requests, large file transfers, embedded objects, and authentication processes. By inspecting packet details, we learned how HTTP headers carry vital information such as status codes, content types, and authorization credentials. The authentication section showed that HTTP Basic Authentication only encodes—rather than encrypts—user credentials, highlighting why secure methods like HTTPS are necessary. Overall, this lab provided valuable hands-on experience in understanding and analyzing real-world HTTP communication.